STUDENT PERFORMANCE PREDICTION

A Predictive Analytics Mini-Project Using Logistic Regression

Course	Predictive Analytics
Project Type	Machine-learning mini-project
Model	Logistic Regression
Dataset	UCI Student Performance — Mathematics dataset
Repository	https://github.com/adityasharmad1d2d3000-afk/student-performance-prediction
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Abstract

This project develops a binary classification model to estimate whether a student passes or fails the mathematics course using demographic, family, school, and study-related attributes. The UCI Student Performance Mathematics dataset is processed using a scikit-learn pipeline, with missing-value imputation, scaling of numerical features, one-hot encoding of categorical features, and logistic regression. The final mathematics grade (G3) is converted into a binary target: pass for G3 ≥ 10 and fail for G3 < 10. The target grade itself is excluded from the input features. On a stratified 80/20 train-test split, the model achieved 87.34% accuracy, an F1-score of 0.900 for the pass class, and ROC-AUC of 0.9485. These results are promising for this split, but further validation is needed before drawing conclusions about performance on other student populations.

1. Introduction

Predictive analytics uses historical data and statistical or machine-learning methods to estimate outcomes for new observations. In education, student data can be examined to identify patterns associated with academic outcomes. This mini-project applies logistic regression to a student performance dataset and evaluates how well it distinguishes students who pass from those who fail.

2. Objectives

- Explore student performance data and inspect its basic structure.
- Prepare numerical and categorical predictors for a machine-learning model.
- Build a logistic regression classifier for pass/fail prediction.
- Evaluate the classifier using accuracy, precision, recall, F1-score, a confusion matrix, and ROC-AUC.
- Document the implementation so it can be reproduced from the GitHub repository.

3. Dataset

The project uses the Student Performance dataset from the UCI Machine Learning Repository, specifically the mathematics file commonly named *student-mat.csv*. The dataset contains 395 student records and 33 columns in the local file used for this project. Attributes include demographic and family information, school-related variables, study habits, absences, and earlier-period grades. The target is derived from G3, the final mathematics grade. A threshold of 10 is used: G3 ≥ 10 is labelled Pass; G3 < 10 is labelled Fail.

Target leakage prevention: after creating the pass/fail label from G3, G3 is removed from the predictor features. This prevents the model from directly using the grade that defines the outcome.

4. Methodology

4.1 Exploratory analysis

The implementation checks the dataset shape and missing values, and saves visualizations of the pass/fail distribution and study time versus final grade. The observed labels in this dataset are 265 Pass and 130 Fail.

4.2 Preprocessing

The data is divided into training and test sets using an 80/20 split, random_state=42, and stratification by the target label. Numerical columns are processed with median imputation and feature scaling. Categorical columns are processed with most-frequent imputation and one-hot encoding. These transformations and the classifier are combined in a scikit-learn pipeline so the preprocessing is fitted on the training data.

4.3 Logistic regression

Logistic regression estimates the probability that an observation belongs to the positive class. In this project, the positive class is Pass. A probability is mapped to a predicted class using the classifier's decision threshold. Logistic regression is suitable for a binary outcome and provides a clear baseline for this prediction task.

4.4 Evaluation

The test set contains 79 records. Performance is reported using accuracy, F1-score for the Pass class, ROC-AUC, a classification report, and a confusion matrix. Accuracy measures the proportion of correct predictions; precision and recall describe the correctness and coverage of positive predictions; F1 is the harmonic mean of precision and recall; ROC-AUC measures ranking discrimination across thresholds.

5. Results

Metric	Observed result	Interpretation
Test accuracy	87.34%	About 87% of test records were classified correctly.
F1-score (Pass class)	0.900	Strong balance of precision and recall for predicting Pass.
ROC-AUC	0.9485	Good class separation on this test split.

Confusion matrix (actual rows × predicted columns; order: Fail, Pass)

	Predicted Fail	Predicted Pass
Actual Fail	24	2
Actual Pass	8	45

The model correctly classified 24 of 26 actual Fail cases and 45 of 53 actual Pass cases. It incorrectly predicted Pass for 2 Fail cases and Fail for 8 Pass cases. The model therefore performs well on this test split, while still making meaningful errors in both directions.

6. Discussion

The reported metrics indicate that logistic regression captures useful patterns in this dataset. The F1-score for Pass is 0.900, and the ROC-AUC of 0.9485 suggests strong ranking discrimination on the held-out split. However, a single random split is not sufficient to establish that the model will perform equally well on new cohorts, different schools, or future academic years. The data is also relatively small, and the chosen pass threshold is a project definition rather than a universal measure of educational success.

7. Limitations and Future Work

- The dataset contains 395 records, and the reported test set contains only 79 records.
- Results are based on one train-test split; stratified cross-validation would provide a more stable estimate.
- The model identifies statistical associations and should not be interpreted as proof of causation.
- Further work could compare logistic regression with LDA/QDA or other syllabus models, examine class-specific errors, and test calibration.
- Any practical use should consider privacy, fairness, context, and the risk of labelling students based on uncertain predictions.

8. Conclusion

This mini-project demonstrates an end-to-end predictive analytics workflow: data inspection, exploratory visualization, preprocessing, model training, and evaluation. Logistic regression achieved 87.34% test accuracy, a Pass-class F1-score of 0.900, and ROC-AUC of 0.9485 on the selected split. The findings show that the dataset contains patterns useful for pass/fail classification, but additional validation is required before applying the model outside this experiment.

References

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